SPAI Exercise 1
Implementing L-Diversity

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Introduction and Objective

Privacy-preserving data publishing aims to reduce the risk that individual records can be re-identified while retaining sufficient utility for subsequent analysis. A commonly used approach is k-anonymity, which requires each record to be indistinguishable from at least k-1 other records with respect to the selected quasi-identifiers (QI). However, k-anonymity alone does not ensure diversity of the sensitive attribute within an equivalence class. An equivalence class may satisfy k-anonymity while all records still share the same sensitive value.

To address this limitation, this project extends an existing Top-down Greedy (TDG) k-anonymization implementation with distinct l-diversity. Distinct l-diversity requires each equivalence class to contain at least l distinct values of the sensitive attribute. The implementation therefore enforces both a minimum equivalence-class size of k and a minimum sensitive-value diversity of l.

The implementation is evaluated on the cahousing dataset using the quasi-identifiers already defined in the original framework and ocean_proximity as the sensitive attribute. The main objective is to investigate how adding l-diversity affects the privacy–utility trade-off compared with the original TDG k-anonymity algorithm. The comparison focuses on three aspects: whether the generated data satisfies the requested privacy constraints, how much information is lost through generalization, and how anonymization affects downstream classification performance.

Accordingly, the project evaluates multiple combinations of k and l and compares them using information-loss metrics such as Normalized Certainty Penalty (NCP), CAVG, and Discernibility Metric (DM), as well as weighted F1 score for classification utility. The goal is not only to demonstrate that l-diversity can be integrated into TDG, but also to quantify the additional utility cost introduced by the stronger privacy requirement.

Dataset and Methodology

The cahousing dataset is already provided in the GitHub repository from the instructions of this exercise (https://github.com/kaylode/k-anonymity/tree/main/data/cahousing). It contains 20.640 records. The original framework already defines five attributes as quasi-identifiers:

- Longitude
- Latitude
- Housing_median_age
- Median_income
- Median_house_value

The attribute ocean_proximity is used as the sensitive attribute and also serves as the target variable for the classification experiments. It contains three distinct values.

The underlying anonymization method is the existing TDG algorithm. TDG recursively partitions the dataset into smaller groups based on distances between records and uses NCP to guide the partitioning process. The original implementation enforces k-anonymity by preventing partitions smaller than k and by applying a balancing procedure whenever a candidate split would otherwise violate the minimum group-size requirement.

To extend TDG with distinct l-diversity, an additional validity check was introduced for each candidate partition. A partition is considered valid only if it contains at least k records and, when l-diversity is enabled, at least l distinct values of the sensitive attribute. Since the TDG implementation internally reorders the dataset so that quasi-identifiers appear first, the position of the sensitive attribute is recalculated dynamically after this reordering.

First implementation

A first implementation rejected a split immediately whenever one of the resulting child partitions failed the l-diversity requirement. This proved too restrictive for k=2 and l=2. The algorithm retained the complete dataset as a single equivalence class, resulting in maximum generalization and very poor utility.

Final implementation

To improve this behavior, the algorithm was modified to try several candidate TDG splits. For each candidate, the original k-anonymity balancing step is applied first, after which both child partitions are checked for k-anonymity and l-diversity. The first valid split is accepted. If no valid split is found after 10 attempts, the parent partition is retained.

Because the original TDG implementation uses randomized pair selection, a fixed seed of 42 was used to make the experiments reproducible. This ensures that repeated runs with the same privacy parameters result in the same partitioning and therefore comparable utility metrics.

Experimental Setup

The experiments compare the original TDG k-anonymity implementation with the extended TDG + l-diversity version. To evaluate how privacy strength affects utility, several combinations of k and l were tested. The k-anonymity baselines use $k \in \{2,5,10\}$, while the l-diverse variants additionally use $l \in \{2,3\}$. Since the sensitive attribute ocean_proximity contains only three distinct values, l=3 represents the strongest meaningful distinct l-diversity setting for this dataset.

Utility

Utility is assessed from two perspectives. First, information loss is measured using the metrics already provided by the anonymization framework:

- Normalized Certainty Penalty (NCP)
- CAVG
- Discernibility Metric (DM)

For NCP and DM, lower values indicate less information loss, while CAVG is best when it is close to 1. The anonymization runs were executed with a fixed random seed of 42 to ensure reproducibility.

Classification Utility

Second, classification utility is evaluated using a Random Forest classifier. The target variable is ocean_proximity, which is also used as the sensitive attribute. The same predefined train/validation split is used for all configurations in order to make the classification scores directly comparable. Performance is measured using the weighted F1 score. For generalized numerical quasi-identifiers, interval values are converted to their midpoint before classification; fully generalized values are represented by the midpoint of the original attribute domain.

Privacy Adherence

In addition to utility, every anonymized output is independently validated for compliance with the requested privacy constraints. Equivalence classes are reconstructed by grouping records according to their generalized quasi-identifiers. For each configuration, the minimum equivalence-class size, the minimum number of distinct sensitive values, and the number of k-anonymity and l-diversity violations are recorded. This makes it possible to confirm that any observed loss in utility results from enforcing the intended privacy guarantees rather than from invalid anonymization output.

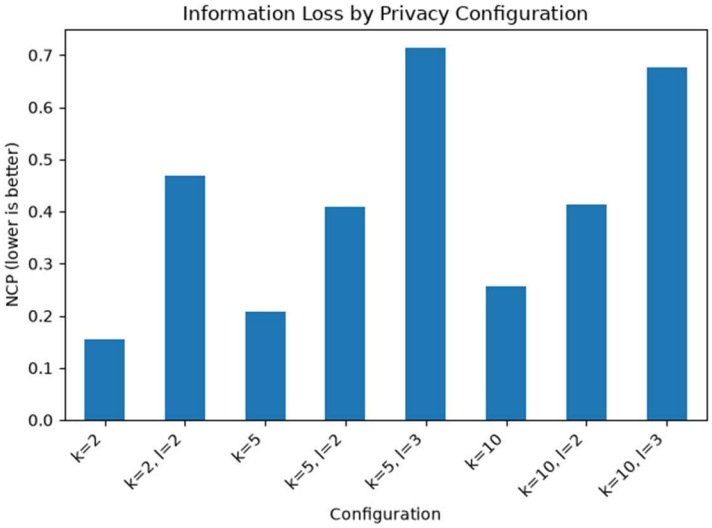
Results

All evaluated anonymized datasets satisfied the requested privacy constraints. No k-anonymity violations were observed, and all l-diverse configurations also had zero l-diversity violations. The minimum equivalence class size always met the configured k, while the minimum number of distinct sensitive values met the configured l for all l-diverse runs.

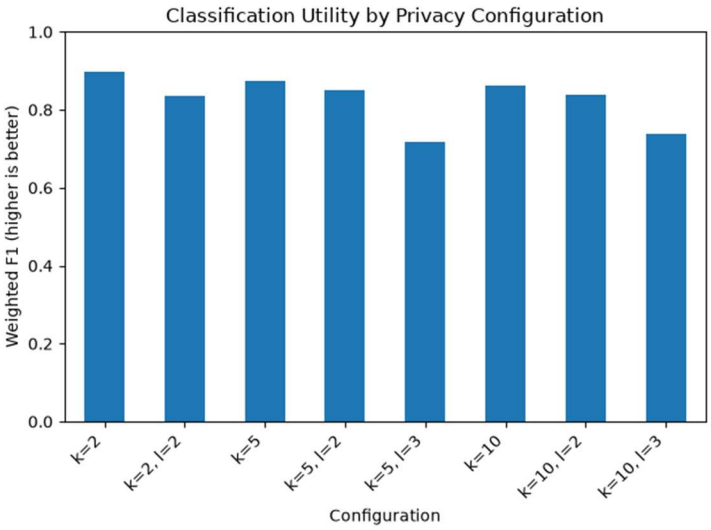
The overall results are summarized in the table below:

k	l	Method	NCP	CAVG	DM	Equivalence Classes	Weighted F1
2	–	TDG k-anonymity	0,155	1,208	56.764	8.544	0,898
2	2	TDG + l-diversity	0,469	15,426	4.360.456	669	0,835
5	–	TDG k-anonymity	0,208	1,499	208.668	2.753	0,874
5	2	TDG + l-diversity	0,408	5,694	2.658.566	725	0,849
5	3	TDG + l-diversity	0,713	28,867	15.679.402	143	0,716
10	–	TDG k-anonymity	0,257	1,502	417.114	1.374	0,861
10	2	TDG + l-diversity	0,414	3,522	2.629.578	586	0,839
10	3	TDG + l-diversity	0,676	14,535	10.982.812	142	0,739

For the non-anonymized original dataset, the Random Forest achieved a weighted F1 score of approximately 0,954. This serves as a reference for the classification utility before anonymization.



The information-loss results show that stronger privacy requirements generally lead to greater generalization. For the original TDG implementation, increasing k from 2 to 10 raises NCP only moderately, from 0,155 to 0,257. In contrast, introducing l-diversity has a much stronger effect. At k=5, NCP increases from 0,208 for k-anonymity to 0,408 for l=2 and 0,713 for l=3. A similar pattern is visible for k=10. This indicates that requiring diversity of the sensitive attribute has a larger impact on information loss than increasing the minimum equivalence-class size alone.



The classification results show a corresponding decrease in predictive utility as the privacy constraints become stronger. The non-anonymized dataset achieves the highest weighted F1 score of approximately 0,954. Under standard k-anonymity, performance decreases gradually from 0,898 at k=2 to 0,861 at k=10. The l-diverse configurations show a larger reduction, particularly for l=3, where weighted F1 falls to 0,716 for k=5 and 0,739 for k=10. These results confirm that stronger privacy protection comes at the cost of reduced classification performance, especially when the diversity requirement is increased.

An interesting result is that utility does not change strictly monotonically with k once l-diversity is enabled. For example, the k=10, l=3 configuration achieves slightly lower NCP and higher weighted F1 than k=5,l=3. This suggests that the interaction between k-anonymity, l-diversity, and the TDG partitioning process depends on the structure of the generated partitions rather than only on the numerical privacy parameters.

Overall, the experiments show that the l-diversity extension successfully enforces the intended privacy constraints while introducing a measurable utility cost. Increasing k alone causes a moderate loss of information, whereas increasing l has a substantially stronger effect on both information-loss metrics and classification performance.

The results therefore confirm a clear privacy-utility trade-off. Stronger diversity requirements improve protection of the sensitive attribute, but require more generalization and reduce predictive utility. The most suitable combination of k and l therefore depends on the acceptable balance between privacy and data utility.

References

kaylode. (archived Sept. 2026). k-anonymity [GitHub repository]. GitHub. https://github.com/kaylode/k-anonymity

Šarčević, T. (2026). FH Technikum Wien. *Security & Privacy in AI – Lecture 1: Data Anonymisation*. course slides.